Brandon Liu

Seattle, WA | bliu8923@gmail.com | linkedin.com/in/bliu8923 | brliu.com

Experience

Amazon Sep 2023 – Present
Software Development Engineer (Java, TypeScript, React, AWS CDK) *Seattle, WA*

- Own development of backend and frontend services for Alexa Shopping Lists handling ~4k requests/sec across Amazon Lists infrastructure.
- Launched new wishlist functionality across ~98% of Amazon product pages and serving hundreds of millions of monthly active users and generating $100M+ in customer sales.
- Reduced production errors in core Alexa Shopping Lists services through reliability improvements and distributed system debugging while supporting on-call operations.
- Designed scalable testing infrastructure for owned services to accelerate validation and deployment cycles.

Amazon Jun 2022 – Sep 2022
Software Development Engineer Intern (Java, AWS CDK, Redis) *Seattle, WA*

- Improved latency of a commonly used Alexa Shopping List API from ~**300ms to 15ms** through caching and architectural changes.
- Built and modified AWS CDK infrastructure packages to streamline provisioning of cloud services for engineering teams.
- Received a return offer for a full-time SDE role.

DeepWater Exploration Inc. Jan 2023 – Jun 2023
Project Management Consultant (Python, OpenCV, YOLOv8) *San Diego, CA*

- Led development of an underwater object detection pipeline using Python, OpenCV, and YOLOv8.
- Built end-to-end computer vision pipeline from dataset labeling to real-time inference for underwater imagery.

Salk Institute Sep 2022 – Jun 2023
Machine Learning Researcher / Data Scientist (PyTorch, GNNs) *San Diego, CA*

- Developed machine learning models and data processing pipelines on the ABCD longitudinal MRI dataset to analyze potential indicators of mental health disorders in children.
- Implemented spatio-temporal graph attention network models using PyTorch.

OrangeSky Jun 2021 – Sep 2021
Software Engineer / Data Analyst Intern (Python, NLP) *San Francisco, CA*

- Built a Python library enabling traders to interact with internal trading platform APIs, including authentication, order execution, and price retrieval.
- Performed sentiment analysis on **15,000** cryptocurrency-related social media posts using VADER and automated a feedback loop for new text data.

Projects

Protect the President Lua, PICO-8
Endless fighting game with progressive difficulty and custom gameplay systems.

Alexa Smart Home Controls Python, C++, JavaScript, AWS, Photon
Built Alexa-powered blinds and TV remote system using servos, IR control, and cloud-connected automation.

IEEE Floating Point Converter ARM Assembly, C
Converted IEEE floating-point values to decimal and hexadecimal on a Raspberry Pi cluster.

Education

University of California San Diego 2019 – 2023
B.S. Data Science, Core GPA: 3.99/4.00 *La Jolla, CA*

Skills

Languages: Java, Python, TypeScript, JavaScript, C++, Lua
Frameworks/Libraries: React, Node.js, PyTorch, TensorFlow, OpenCV
Cloud/Tools: AWS, AWS CDK, Docker, Git